

NABEEL AHMED KABIR

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Education

York University

Expected Graduation: April 2027

Bachelor of Computer Engineering

Toronto

- Courses: Data Structures and Algorithms, Design and Analysis of Algorithms, Software Design, Operating System Fundamentals, Computer Networks / Network Protocols, Computer Organization, Digital Logic Design

Work Experience

Cotriply

Apr 2026 – Jun 2026

AI Software Developer Intern

Toronto, Canada

- Transitioning an **AI-powered group travel coordination platform** from prototype to MVP as part of the **Advance Ontario** program
- Refactored **16+ monolithic files** (some **6,000+ lines**) into modular **TypeScript** components, reducing codebase size by **50%** with zero errors
- Built an automated **email notification system** covering all core trip events (document uploads, polls, itinerary updates) for all trip members
- Configured a **Vercel** staging environment integrated with **GitHub**, decoupling development from the live production platform
- Resolved **10+ bugs** across trip creation, UI overflow, poll logging, and itinerary auto-save through **Postman** API testing

Projects

Comms-Denied Autonomy Hackathon | 3rd Place, \$1000 – Red Team DefTech

Python | OpenCV

- Built a fully autonomous **drone navigation system** in **Python** using **OpenCV** to interpret visual cues for real-time flight decisions with **zero human intervention or GPS**
- Designed a **computer vision pipeline** using **HSV color masking** and **contour analysis** to detect and classify objects, replacing an unreliable LLM-based approach for improved speed and reliability
- Collaborated with a **5-person team** under a **9-hour constraint**, owning the vision pipeline and decision-logic components; placed **3rd of 10 teams**

FPGA Snake Game | Demo | Source Code

Verilog | Quartus Prime | DE10-Lite

- Implemented **Snake** entirely in hardware on an **FPGA** using a multi-state **Finite State Machine (FSM)** in **Verilog**, with **VGA signal generation** and clock division driving real-time rendering
- Built collision detection (walls and self) and score tracking logic, collaborating with a partner to resolve hardware input reliability issues through debouncing and timing control

ISS Live Tracker | Source Code

JavaScript | HTML/CSS | Leaflet.js

- Built a real-time web app using **JavaScript**, **HTML/CSS**, and **Leaflet.js** that tracks the ISS live on an interactive map, pulling live position data from the **WhereTheISS.at API** and refreshing every **5 seconds**
- Added interactive features including **auto-centering** ("Follow ISS"), **orbit trail history**, a visibility radius overlay, and a **NASA GIBS night-lights layer** as modular front-end components

Evolutionary AI Racer | Source Code

Python | Pygame | Neural Networks

- Built an evolutionary AI racing simulator in **Python** and **Pygame** where **neural-network-controlled** agents learn steering and movement through a **genetic algorithm** (selection, crossover, mutation)
- Improved lap completion rate from near **0% to 70%+** over **300 generations**, reducing iteration time by **50%** through modular, replay-driven debugging

Technical Skills

Languages: Python, JavaScript, TypeScript, Java, C, SQL, HTML5/CSS3, Bash, Verilog

Developer Tools: Git, GitHub, VS Code, Postman, Microsoft Azure, Vercel, Linux/Unix, Quartus Prime

Libraries/Frameworks: React, OpenCV, pandas, NumPy, scikit-learn, Leaflet.js, MySQL, MongoDB